



KIT - KALAI GNAN KARUNANIDHI INSTITUTE OF TECHNOLOGY

(An Autonomous Institution, Affiliated with Anna University, Chennai and Approved
by AICTE, New Delhi.) Coimbatore - 641 402, Tamil Nadu, India.

SPRINTATHON

SCHOOL OF COMPUTING

DEPARTMENT OF

COMPUTER SCIENCE AND ENGINEERING

&

CSE (CYBER SECURITY)

HOSTED ON



We are fully
engaged in
carrying out the



TEAM OVERVIEW

Problem Statement Id :-PS26SAM037

PS Category :-PS 7. Real-Time

Fraud Anomaly Detector

Theme :-FinTech + Cybersecurity

Team Name :- LUCK -404

Team Members :-

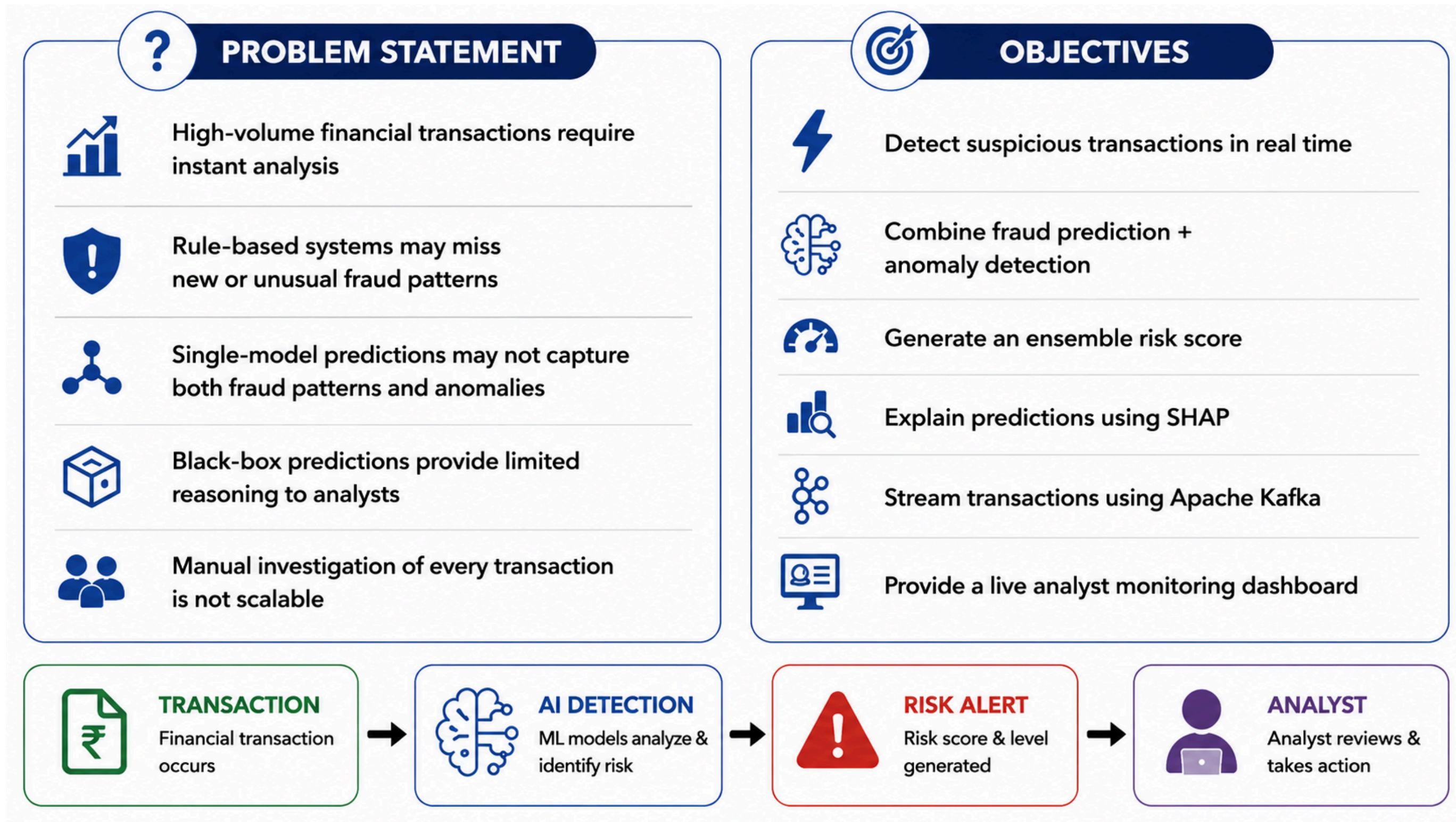
1.GNANA PRAKASH D -II AI &ML

2.SRIRAM M -II AI&ML

3.KISHOR RAJ V -II AI&ML

4.KELVIN IMMANUVEL -II AI&ML

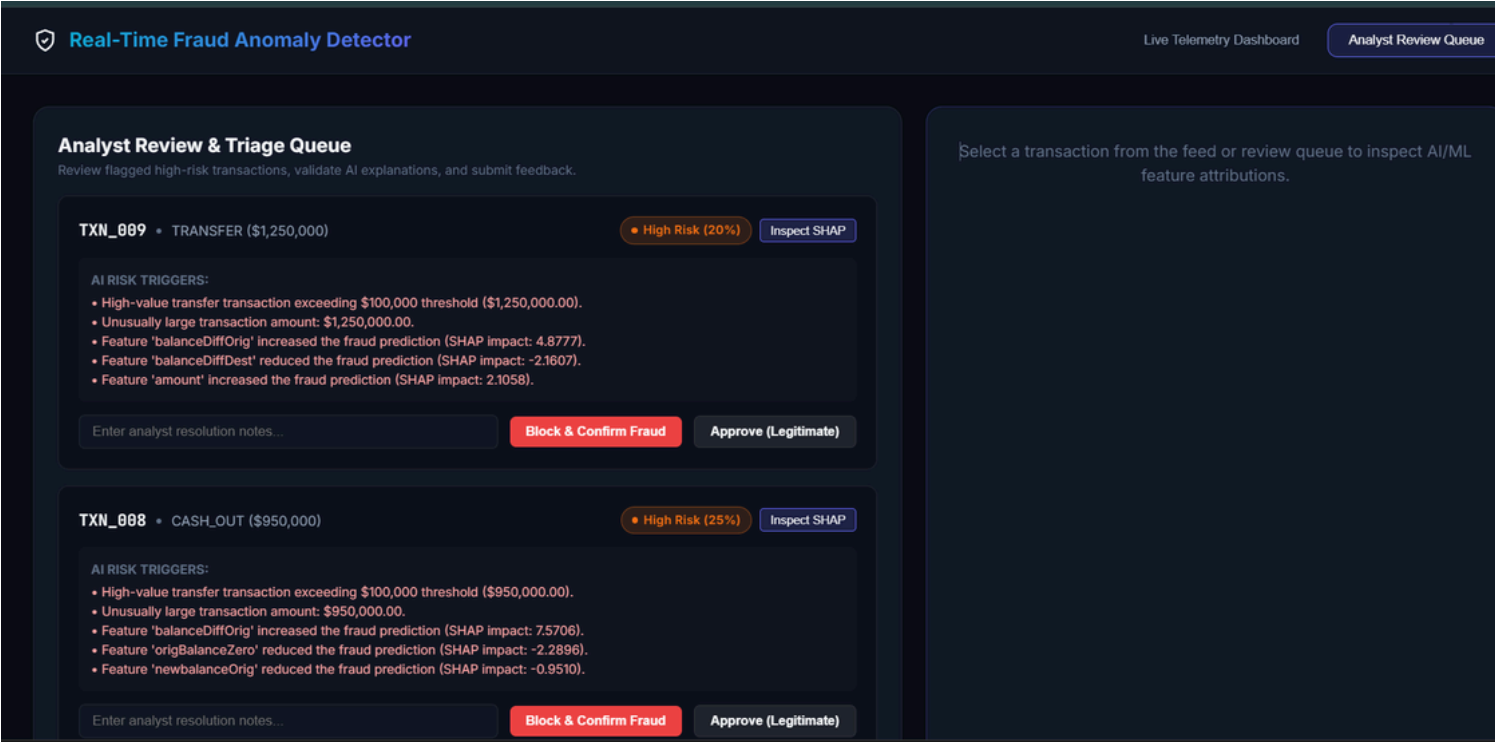
PROBLEM STATEMENT & OBJECTIVES





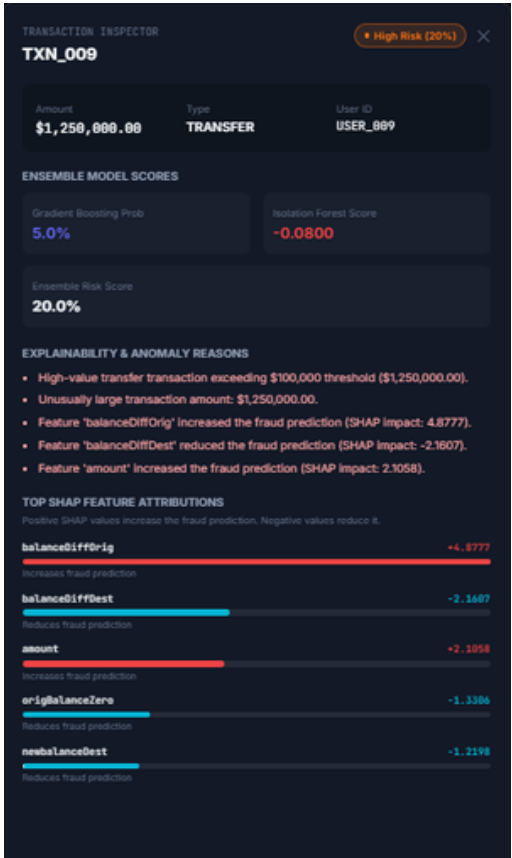
PROPOSED SOLUTION

Existing Limitations	Our Solution
 Fixed-rule dependency	 ML-based fraud prediction
 Known fraud pattern only	 Isolation Forest anomaly detection
 Single-model dependency	 Ensemble scoring
 Batch / late processing	 Kafka real-time streaming
 Black-box prediction	 SHAP explainability
 Raw backend output	 React analyst dashboard
 No persistent history	 PostgreSQL
 Hardcoded performance claims	 Real evaluation metrics
 Poor operational visibility	 TPS + latency + percentile metrics
 Manual prioritization difficult	 Risk levels + review queue

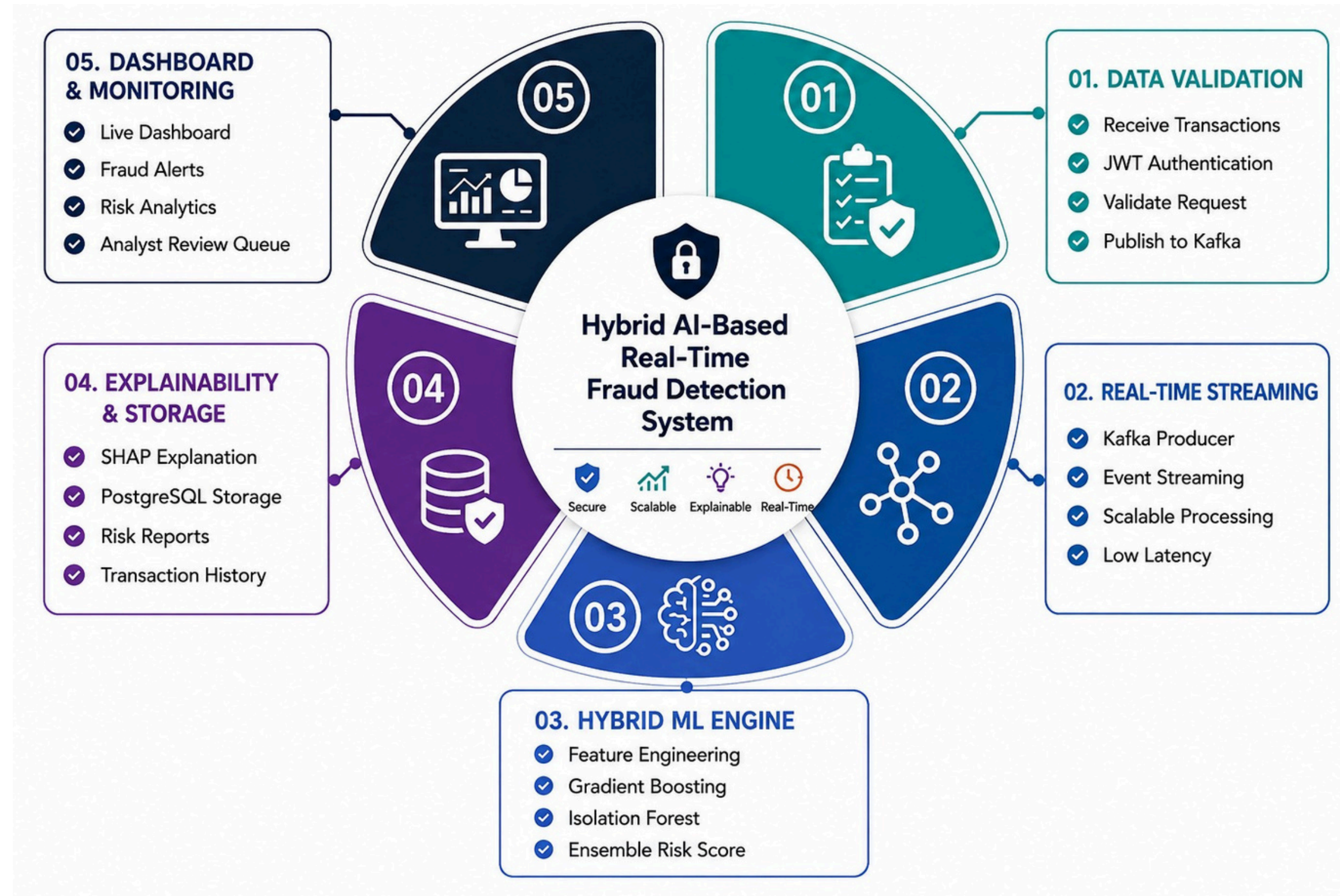


ANALYST REVIEW & TRIAGE QUEUE

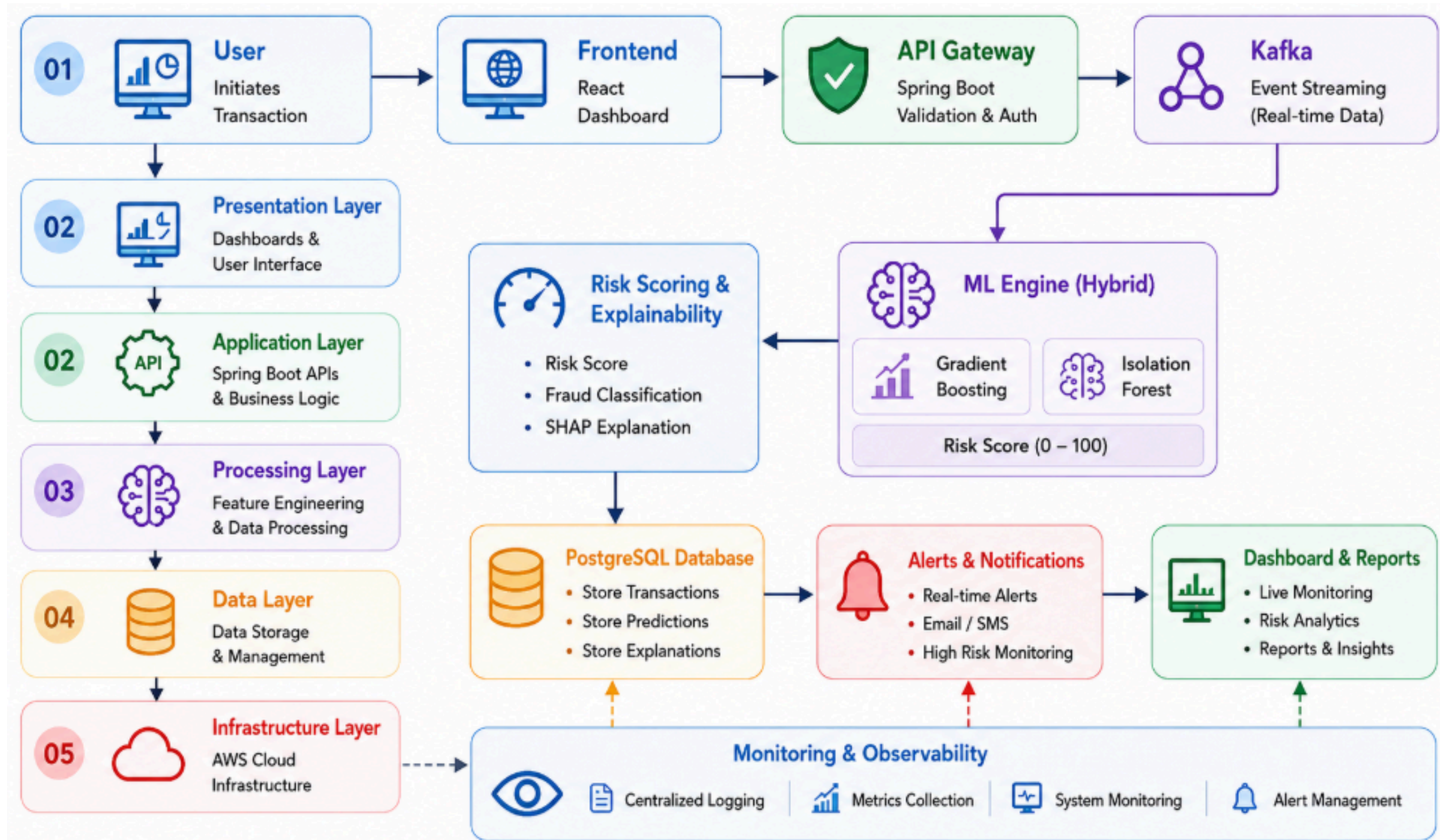
TRANSACTION INSPECTOR



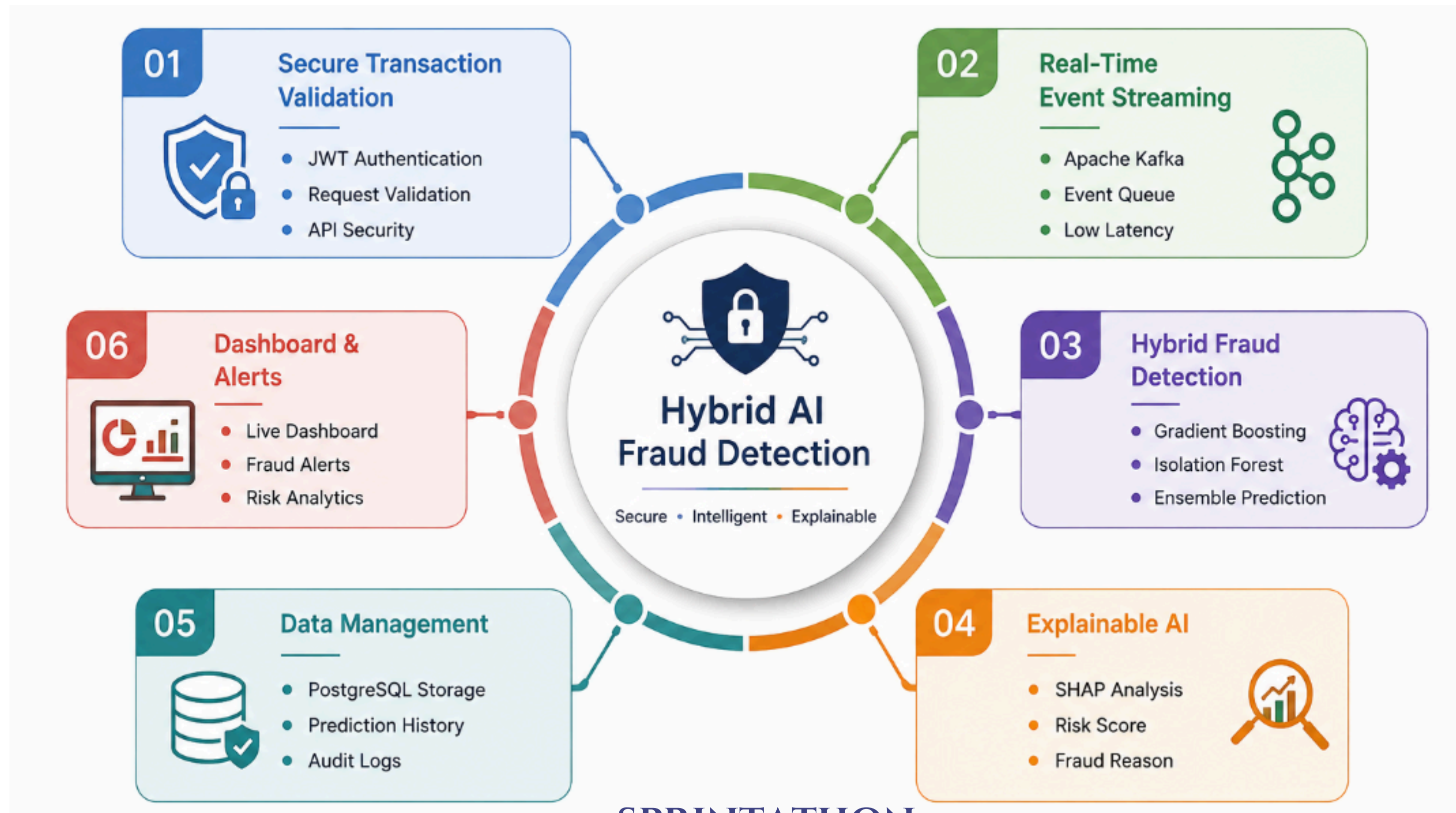
TECHNICAL APPROACH



ARCHITECTURAL BREAKDOWN & BLOCK DIAGRAM



CORE FUNCTIONAL UNITS





VALIDATION & EMPIRICAL FINDINGS

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=== Gradient Boosting Results ===

Fraud Precision: 0.9704 (97.04%)
Fraud Recall: 0.9379 (93.79%)
Fraud F1 Score: 0.9539 (95.39%)

Classification Report:
      precision    recall  f1-score   support

     0       0.99       1.00       1.00      18357
     1       0.97       0.94       0.95       1643

 accuracy          0.99          0.99      20000
  macro avg       0.98       0.97       0.97      20000
 weighted avg     0.99       0.99       0.99      20000

Confusion Matrix:
[[18310   47]
 [  102 1541]]

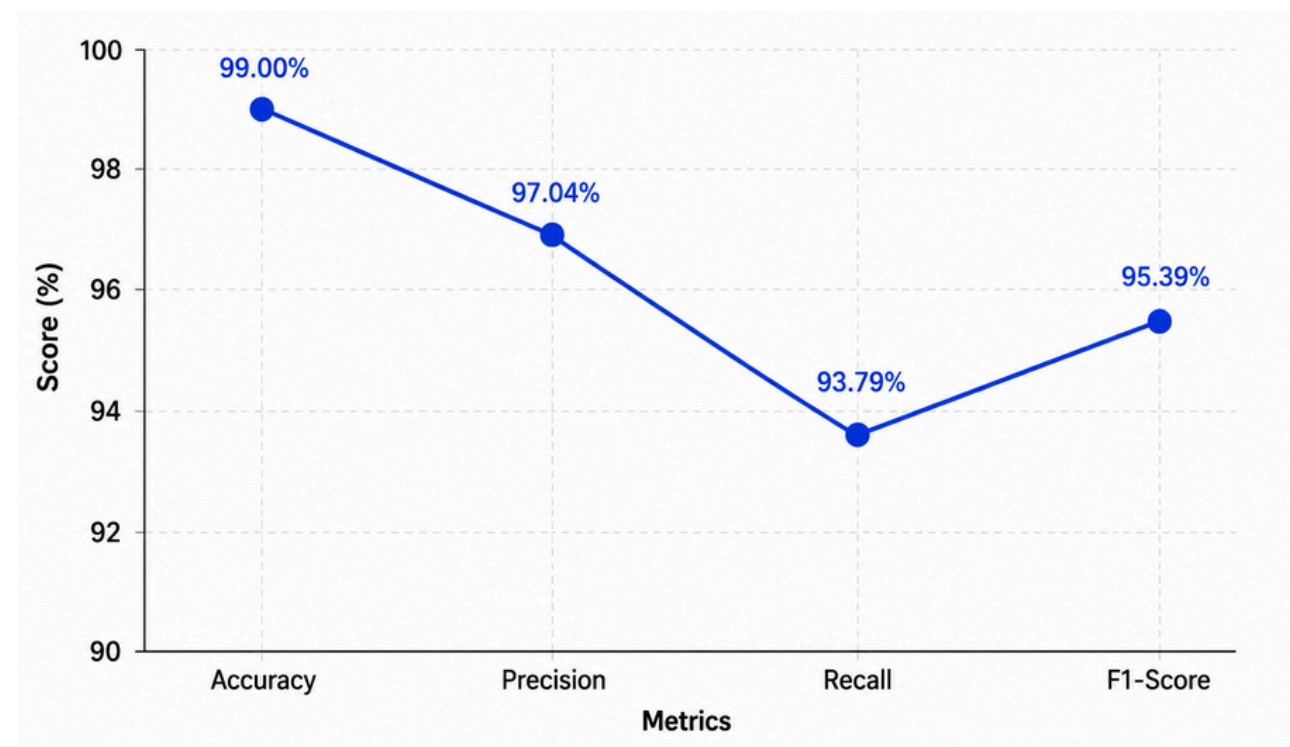
Confusion Matrix Details:
True Negatives : 18310
False Positives: 47
False Negatives: 102
True Positives : 1541
```

DATASET SPLIT

Total Dataset : 100,000

Training Set : 80,000 (80%)

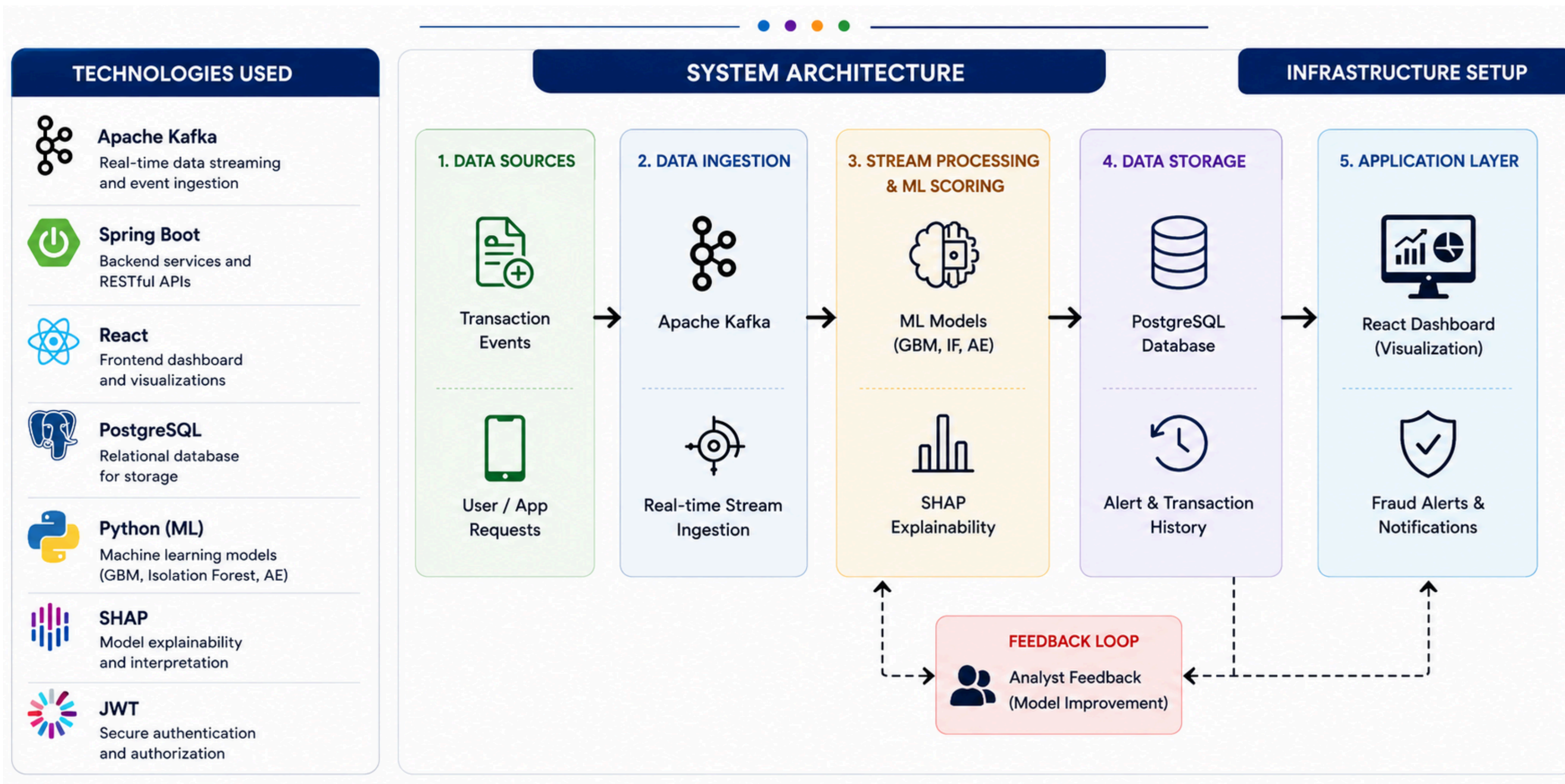
Testing Set : 20,000 (20%)



CLASSIFICATION REPORT

Metric	Value
Precision	97.04%
Recall	93.79%
F1-Score	95.39%
Support	20,000

RESOURCE & INFRASTRUCTURE SETUP





DELIVERABLES

CATEGORY	DELIVERABLE
REAL-TIME STREAMING	APACHE KAFKA TRANSACTION STREAMING PIPELINE
BACKEND SYSTEM	SPRING BOOT REST API & BUSINESS LOGIC
HYBRID ML ENGINE	GRADIENT BOOSTING + ISOLATION FOREST + AUTOENCODER
EXPLAINABLE AI	SHAP-BASED FRAUD EXPLANATIONS
DATABASE	POSTGRESQL TRANSACTION & FEEDBACK STORAGE
FRONTEND	REACT LIVE MONITORING DASHBOARD
ANALYST MODULE	REVIEW QUEUE & FRAUD VERIFICATION
MONITORING	LIVE TPS, LATENCY & MODEL METRICS DASHBOARD
DEPLOYMENT	DOCKERIZED MULTI-SERVICE APPLICATION

PERFORMANCE

REAL-TIME FRAUD ANOMALY DETECTOR – PERFORMANCE
MODEL ACCURACY
PRECISION (FRAUD CLASS) : 97.04%
RECALL (FRAUD CLASS) : 93.79%
F1-SCORE : 95.39%
ROC-AUC (ENSEMBLE) : 0.997
LATENCY SLA
P50 INFERENCE LATENCY : 2.1 ms ✓ SLA < 5ms
P95 INFERENCE LATENCY : 3.5 ms ✓ SLA < 7ms
P99 INFERENCE LATENCY : 4.2 ms ✓ SLA < 10ms
REDIS CACHE LOOKUP : < 1.0 ms
THROUGHPUT
SINGLE REPLICA : 1,500 TPS
DUAL REPLICA (K8s) : ~3,000 TPS
FALSE POSITIVE RATE : 0.26% (vs >85% RULES)
FRAUD CAUGHT (TP) : 1,541 / 1,643 (93.8%)

Applications



01 — Banking & Digital Payments

- Detect suspicious transactions
- Prevent unauthorized transfers

01



02 — UPI & Mobile Wallets

- Secure online payments
- Detect suspicious activity

02



03 — E-Commerce & FinTech Platforms

- Reduce payment fraud
- Protect customers & merchants

03



04 — Fraud Monitoring Dashboard

- Real-time fraud alerts
- Analyst review & risk monitoring

04



References

RESEARCH PAPERS

- Fukas, P. et al. (2022)
Towards Explainable Artificial Intelligence in Financial Fraud Detection: Using Shapley Additive Explanations to Explore Feature Importance
DOI: https://doi.org/10.1007/978-3-031-07472-1_7
- Almalki, F., & Masud, M. (2025) Financial Fraud Detection Using Explainable AI and Stacking Ensemble Methods
arXiv: <https://arxiv.org/abs/2505.10050>
DOI: <https://doi.org/10.48550/arXiv.2505.10050>
- Explainable Artificial Intelligence and Causal Inference based ATM Fraud Detection
arXiv: <https://arxiv.org/abs/2211.10595>

OFFICIAL DOCUMENTATION

- Apache Kafka Documentation
- Spring Boot Documentation
- FastAPI Documentation
- React Documentation
- PostgreSQL Documentation
- SHAP Documentation
- Scikit-learn Documentation

DATASET

- [PaySim – Mobile Money Fraud Simulation Dataset \(Kaggle\)](#)